Operations Research: the five-minute formula sheet

Lectures 1–5. Every line is a formula or a rule you may quote, with what it means beside it. Conventions of the course: **maximisation**, equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$, A is $m \times n$ with $\operatorname{rank} A = m$, tableau written *equation style* (basic variables on the left). Last revised 2026-11-03.

1. Transformations to equational form

$a^\top x \leq b \ \Rightarrow\ a^\top x + s = b,\ s \geq 0$	slack. The all-slack basis is feasible only if $b \geq 0$.
$a^\top x \geq b \ \Rightarrow\ a^\top x - e = b,\ e \geq 0$	surplus (coefficient -1). All-surplus basis gives $e = -b < 0$: not a feasible start.
$a^\top x = b \ \Rightarrow$ nothing added	an equality row needs no new variable; adding one in either direction is wrong.
$\min c^\top x \ \Rightarrow\ \max(-c)^\top x, \quad \text{cost} = -\text{value}$	report the original cost; multipliers keep their meaning up to the sign flip.
$x \geq L \ \Rightarrow\ x = L + u,\ u \geq 0$	substitute everywhere; a constant appears in z and in the right-hand sides.
$x \leq U \ \Rightarrow\ x = U - u,\ u \geq 0$	same, with the sign of every coefficient of x flipped.
$L \leq x \leq U \ \Rightarrow\ x = L + u, \quad u + s = U - L$	one shifted variable plus one row (fewest variables).
x free $\Rightarrow\ x = u - v,\ u, v \geq 0$	or eliminate x with an equality row; u and v are never both basic.
$\min \max_i \ell_i(x) \ \Rightarrow\ \min t, \quad t \geq \ell_i(x)\ \forall i$	at an optimum t equals the largest piece. Maximin: $\max t,\ t \leq \ell_i(x)$.
$\lvert\ell(x)\rvert \leq \varepsilon \ \Rightarrow\ \ell(x) \leq \varepsilon,\ -\ell(x) \leq \varepsilon$	two rows. $\lvert\ell(x)\rvert \geq \varepsilon$ is *not* linear (non-convex).
$\min \sum_i \lvert e_i\rvert \ \Rightarrow\ e_i = u_i - v_i,\ \min \sum(u_i + v_i)$	at an optimum one of u_i, v_i is zero, so $u_i + v_i = \lvert e_i\rvert$.
$\frac{x_1}{x_1+x_2} \geq \alpha \ \Rightarrow\ (1-\alpha)x_1 - \alpha x_2 \geq 0$	clear the denominator (positive); then treat as a $\geq$ row.
$b_i < 0$: multiply the row by -1	flips $\leq$ into $\geq$; then slack becomes surplus and the slack start fails.
$z \to \lambda z\ (\lambda > 0)$: $x^\star$ same, $y \to \lambda y$; $b \to \lambda b$: $x^\star \to \lambda x^\star$, y same	scaling the objective scales the shadow prices; scaling all resources scales the plan.
n = #vars after splits+#slacks+#surpluses, m = #rows	count n and m *after* the transformation; equality rows add to m only.

2. Bases, basic solutions, vertices

$B = [A_{B(1)} \cdots A_{B(m)}],\ \det B \neq 0; \quad x_B = B^{-1}b,\ x_N = 0$	a basis is m independent columns; the basic solution sets the other $n - m$ variables to 0.
$\#\text{bases} \leq \binom{n}{m}$	upper bound only: some m-subsets are singular.
basic feasible solution $\iff B^{-1}b \geq 0$	feasible basic solutions are exactly the vertices (extreme points) of the polyhedron.
degenerate $\iff$ some $x_{B(k)} = 0$	a degenerate vertex has several bases; nondegenerate $\iff$ every basic variable > 0.
Fundamental theorem: feasible and bounded $\Rightarrow$ an optimal vertex exists	so the simplex may restrict itself to basic feasible solutions.
at most m positive variables in a basic solution	a solution with more than m positive variables is not basic.

3. From a basis B to the tableau (equation style)

$x_B = \beta + Q\,x_N, \quad \beta = B^{-1}b, \quad Q = -B^{-1}A_N$	basic variables on the left, nonbasic on the right; a displayed column is $-B^{-1}A_j$.
$y^\top = c_B^\top B^{-1}$ (lecture: $p^\top = c_B^\top B^{-1}$)	multipliers = shadow prices = dual solution. One per row.
$r_j = c_j - y^\top A_j \quad (r_B = 0)$	reduced cost: gain in z per unit of x_j brought in.
$z = z_0 + \sum_{j \in N} r_j x_j, \qquad z_0 = c_B^\top \beta = y^\top b$	the z row; the constant is the current value.
column of slack $s_i \ = \ -B^{-1}e_i \ = \ -(\text{column } i \text{ of } B^{-1})$	read B^{-1} off the slack columns, sign flipped; a basic slack shows a unit column.

$r_{s_i} = -y_i \quad \Rightarrow \quad y_i = -(\text{reduced cost of } s_i)$	the dual solution is the negative of the slack reduced costs.
optimal $\iff \beta \geq 0$ and $r \leq 0$ (max)	primal feasibility and dual feasibility together. Both tests are what sensitivity analysis re-checks.

4. Simplex (maximisation)

entering j : $r_j > 0$ (Bland: smallest such j)	any positive reduced cost improves z; Bland's rule prevents cycling.
ratio test: $\theta = \min_{k:\, q_{kj}<0} \frac{\beta_k}{-q_{kj}}$, leaving $=$ argmin	the first basic variable to hit 0 leaves (Bland: smallest index among ties).
no $q_{kj} < 0 \Rightarrow$ unbounded	x_j can grow forever, $z \to +\infty$; the ray is $x_B = \beta + Q_j t$, $x_j = t$.
$\theta = 0 \Rightarrow$ degenerate pivot	basis changes, vertex and z do not.
optimal, some nonbasic $r_j = 0$ with a finite ratio $\Rightarrow$ alternative optima	pivot on it: another optimal vertex; the segment between them is optimal.
optimal, nondegenerate, all nonbasic $r_j < 0 \Rightarrow$ unique optimum	the converse of the optimality test holds only under nondegeneracy.
dual simplex: row with $\beta_k < 0$, enter j with $q_{kj} > 0$ minimising $\frac{-r_j}{q_{kj}}$	keeps $r \leq 0$ while repairing $\beta < 0$. Use after a right-hand-side change leaves the range.
no $q_{kj} > 0$ in that row $\Rightarrow$ primal infeasible	the row says $x_{B(k)} = \beta_k + (\text{nonpositive}) < 0$.

5. Duality

$\max\{c^\top x : Ax \leq b,\ x \geq 0\} \iff \min\{b^\top y : A^\top y \geq c,\ y \geq 0\}$	canonical pair; one dual variable per primal row, one dual row per primal variable.
Sign rules, primal max**:** row $\leq \Rightarrow y_i \geq 0$; row $\geq \Rightarrow y_i \leq 0$; row $= \Rightarrow y_i$ free; $x_j \geq 0 \Rightarrow$ dual row $\geq c_j$; $x_j \leq 0 \Rightarrow \leq c_j$; x_j free $\Rightarrow = c_j$.	for a primal min every inequality flips: row $\geq \Rightarrow y_i \geq 0$, row $\leq \Rightarrow y_i \leq 0$, $x_j \geq 0 \Rightarrow$ dual row $\leq c_j$, $x_j \leq 0 \Rightarrow \geq c_j$. Check by converting to canonical form.
weak duality: $c^\top x \leq b^\top y$	for every primal feasible x and dual feasible y (max primal); proof: $c^\top x \leq (A^\top y)^\top x = y^\top Ax \leq y^\top b$.
strong duality: $c^\top x^\star = b^\top y^\star$	if one side has an optimum, so does the other, same value; $y^\star =$ the multipliers of the optimal basis.
complementary slackness: $y_i(b_i - A_i x) = 0$, $x_j(A_j^\top y - c_j) = 0$	feasible x, y are both optimal $\iff$ CS holds. A used-up row may have $y_i > 0$; a slack row has $y_i = 0$.
certificate: y dual feasible and $b^\top y = c^\top x \Rightarrow x$ optimal	how to prove optimality without a tableau.
dual of the dual $=$ primal	so every statement is symmetric.
Outcomes. primal unbounded $\Rightarrow$ dual infeasible; dual unbounded $\Rightarrow$ primal infeasible; both infeasible is possible.	"primal infeasible $\Rightarrow$ dual unbounded" is *false*: the dual may also be infeasible.
$y_i = \frac{\partial z^\star}{\partial b_i}$ (within the range)	shadow price: value of one more unit of resource i; 0 when the resource is not binding (CS).

6. Sensitivity analysis: the one idea

The basis stays optimal as long as both tests still pass: primal feasibility $B^{-1}b \geq 0$ and dual feasibility $r = c - A^\top y \leq 0$.	a change in b touches only the first test; a change in c touches only the second; a new column touches only its own reduced cost; a new row touches only feasibility.
Case 1: resource $b_i \to b_i + \delta$	
$g = B^{-1}e_i = -(\text{column of } s_i \text{ in the tableau})$	how each basic variable moves per unit of δ.
$\beta + \delta g \geq 0 \Rightarrow \max_{g_k>0} \frac{-\beta_k}{g_k} \leq \delta \leq \min_{g_k<0} \frac{-\beta_k}{g_k}$	range of δ keeping the basis optimal; the plan is $x_B = \beta + \delta g$.
$z^\star(\delta) = z^\star + y_i\delta$	value moves at the shadow price, inside the range only.
δ outside the range $\Rightarrow$ dual simplex from $\beta + \delta g$	the z row is still dual feasible; repair the negative entry, then read the new y.
$v(b)$ is concave, piecewise linear, slopes y	shadow prices can only fall as a resource grows; so $v(b + \delta e_i) \leq v(b) + y_i\delta$ for *all* δ.

Case 2: profit $c_j \to c_j + \varepsilon$

x_j nonbasic: $r_j + \varepsilon \le 0 \iff \varepsilon \le -r_j$	plan and value unchanged below the break-even; at $\varepsilon = -r_j$ an alternative optimum appears.
$x_{B(k)}$ basic: add $\varepsilon \times$ (row of $x_{B(k)}$) to the z row	new reduced costs $r_j + \varepsilon q_{kj}$ for every nonbasic j; new value $z^\star + \varepsilon\beta_k$.
$r_j + \varepsilon q_{kj} \le 0\ \forall j \in N \ \Rightarrow$ range of ε	plan unchanged in the range; outside, continue the primal simplex from the current tableau.
$v(c)$ is convex, piecewise linear	so $v(c + \varepsilon e_j) \ge v(c) + \varepsilon x_j^\star$ for *all* ε (old plan still feasible).

Case 3: new activity with profit γ and column a

$r_u = \gamma - y^\top a$, displayed column $= -B^{-1}a$	$r_u \le 0$: do not produce, nothing changes. $r_u > 0$: add the column and pivot it in.
$\gamma \le y^\top a$ (resources valued at shadow prices) $\iff$ not worth it	break-even profit of the new product is $y^\top a$.

New constraint $a^\top x \le \beta$

$a^\top x^\star \le \beta \ \Rightarrow$ nothing changes	otherwise add the row with its slack (basic, negative) and run the dual simplex.

Simultaneous changes

$\beta + \delta g^{(i)} + \delta' g^{(k)} \ge 0$ (two resources)	a polygon in (δ, δ'), not the product of the two ranges; value $z^\star + y_i\delta + y_k\delta'$.
$b_i + \delta,\ c_{B(k)} + \varepsilon:\quad z = z^\star + y_i\delta + \varepsilon\beta_k + \varepsilon g_k\delta$	the cross term $\varepsilon g_k\delta$ comes from the basic variable moving with δ; valid while both tests pass.

Value stability (the σ bound)

$\lvert c'^\top x - c^\top x\rvert \le \sigma\ \forall x$ feasible $\ \Rightarrow\ v_0 - \sigma \le v \le v_0 + \sigma$	lower: old optimum is still feasible, $v \ge c'^\top x_0 \ge v_0 - \sigma$; upper: the new optimum was feasible before, $v = c'^\top x' \le c^\top x' + \sigma \le v_0 + \sigma$.
right-hand-side version: $\lvert b_i' - b_i\rvert \le \sigma_i \ \Rightarrow\ \lvert v' - v\rvert \le \sum_i y_i\sigma_i$ (basis unchanged)	via $z = y^\top b$ while the basis stays optimal; in general use weak duality with the same dual-feasible y.

7. Reading a final tableau in sixty seconds

Plan: basic variables = constants, nonbasic = 0. **Value:** constant of the z row. **Shadow prices:** $y_i = -r_{s_i}$. B^{-1}**:** minus the slack columns. **Binding rows:** s_i nonbasic. **Unique:** all nonbasic $r_j < 0$ and nondegenerate. **Degenerate:** some basic constant 0.

Range of b_i: $\beta + \delta g \ge 0$ with $g = -$(column of s_i). **Range of basic c_k:** $r_j + \varepsilon q_{kj} \le 0$. **Range of nonbasic c_j:** $\varepsilon \le -r_j$. **New product:** $\gamma - y^\top a$. **Check:** $z = y^\top b = c^\top x$.

8. Traps that cost points

Surplus has coefficient -1, not $+1$. An equality row gets no variable. min becomes max$(-c)$ and the answer is minus the value. The slack basis is infeasible when some $b_i < 0$. Optimality for max is $r \le 0$, for min it is $r \ge 0$ (or convert first).

Shadow price of a non-binding row is 0. A $\ge$ row in a max gives $y_i \le 0$. Sensitivity formulas hold *only inside the range*; beyond it, pivot. Degenerate optimum: several bases, several y, and "all $r_j < 0$" no longer proves uniqueness. Both primal and dual can be infeasible.